

HDOC – 18th day

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Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

PROBLEM STATEMENT:

Write a menu-driven C program using switch-case to perform digit-based operations on a positive integer. Define one separate user-defined function for each menu option, *with void return type* and *with a simple function call (without call by value/reference)*. Each function should read the required input, process it, and display the result.

- Check whether it is a Strong number.
- Check whether it is an Armstrong number.
- Check whether it is a Harshad number.
- Check whether it is a Duck number.

Algorithm:

- 1) Start
- 2) Define user defined database with void return type:
Strong(), Armstrong(), harshad(), duck()
- 3) Declare choice(integer) in int main.
- 4) Display the menu:
 - 1: Check whether it is a Strong number.
 - 2: Check whether it is an Armstrong number.
 - 3: Check whether it is a Harshad number.
 - 4: Check whether it is a Duck number.

5) Read the user choice.

6) Use switch(choice) to perform the operation.

7) If choice = 1, call strong():

7.1) Read a positive integer x.

7.2) Store x in temporary variable tvar.

7.3) Initialize sum = 0.

7.4) Extract the last digit using $\text{digit} = \text{tvar} \% 10$.

7.5) Find the factorial of the digit using a for loop.

7.6) Add the factorial to sum.

7.7) Remove the last digit using $\text{tvar} = \text{tvar} / 10$.

7.8) Repeat until tvar becomes 0.

7.9) If $\text{sum} == \text{x}$, display "Strong Number"; otherwise display "Not a Strong Number".

8) If choice = 2, call armstrong():

8.1) Read a 3-digit positive integer a.

8.2) Store a in tvar.

8.3) Initialize sum = 0.

8.4) Extract each digit using $\text{digit} = \text{tvar} \% 10$.

8.5) Find the cube of the digit using $\text{digit} * \text{digit} * \text{digit}$.

8.6) Add the cube to sum.

8.7) Remove the last digit using $\text{tvar} = \text{tvar} / 10$.

8.8) Repeat until tvar becomes 0.

8.9) If $\text{sum} == \text{a}$, display "Armstrong Number"; otherwise display "Not an Armstrong Number".

9) If choice = 3, call harshad():

9.1) Read a positive integer b.

9.2) Store b in tvar.

9.3) Initialize sum = 0.

9.4) Extract each digit using $\text{digit} = \text{tvar} \% 10$.

9.5) Add the digit to sum using $\text{sum} = \text{sum} + \text{digit}$.

9.6) Remove the last digit using $\text{tvar} = \text{tvar} / 10$.

9.7) Repeat until tvar becomes 0.

9.8) Check $b \% \text{sum}$.

9.9) If $b \% \text{sum} == 0$, display "Harshad Number"; otherwise display "Not a Harshad Number".

Step 10: If choice = 4, call duck():

10.1) Read a positive integer n.

10.2) Store n in tvar.

10.3) Initialize check = 0.

10.4) Extract each digit using $\text{digit} = \text{tvar} \% 10$.

10.5) If $\text{digit} == 0$, set check = 1.

10.6) Remove the last digit using $\text{tvar} = \text{tvar} / 10$.

10.7) Repeat until tvar becomes 0.

10.8) If check == 1, display "Duck Number"; otherwise display "Not a Duck Number".

Step 11: If the user enters any choice other than 1, 2, 3, or 4, display "Invalid choice".

Step 12: Stop.

Test Case Table:

Test Case	Choice	Input	Expected Output	Actual output	Result
1.	1	145	145 is a strong number	145 is a strong number	Pass
2.	1	123	123 is not a strong number	123 is not a strong number	Pass
3.	2	153	It is Armstrong number.	It is Armstrong number.	Pass
4.	2	123	It is not Armstrong number.	It is not Armstrong number.	Pass
5.	3	18	18 is harshad number	18 is harshad number	Pass
6.	3	19	19 is not harshad number.	19 is not harshad number.	Pass
7,	4	1025	1025 is a duck number.	1025 is a duck number.	Pass
8.	4	1234	1234 is not a duck number	1234 is not a duck number.	Pass
9.	5	_____	Invalid choice	Invalid choice	Pass

C Program:

A screenshot of a terminal window titled "kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox". The window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". Below the menu bar is a toolbar with icons for file operations and a tab bar showing tabs 1, 2, 3, and 4. The main area of the terminal displays a C program. The program includes `<stdio.h>` and defines four functions: `strong()`, `armstrong()`, `harshad()`, and `duck()`. The `main()` function starts with a `choice` variable, prints a menu, and uses a `switch` statement to call the appropriate function based on the user's input. The menu options are: 1: Check whether it is a Strong number, 2: Check whether it is an Armstrong number, 3: Check whether it is a Harshad number, and 4: Check whether it is a Duck number. The program prompts the user to "Enter your choice:" and reads the input into `choice`. The `switch` statement has cases for 1, 2, 3, and 4, each calling its respective function and then `break`. A `default` case prints "Invalid choice".

```
1#include<stdio.h>
2void strong();
3void armstrong();
4void harshad();
5void duck();
6
7int main()
8
9{
10    int choice;
11
12    printf("\n MENU \n");
13    printf("1:Check whether it is a Strong number\n\
142:Check whether it is an Armstrong number\n\
153:Check whether it is a Harshad number.\n\
164:Check whether it is a Duck number.\n");
17    printf("Enter your choice:");
18    scanf("%d", &choice);
19
20    switch(choice)
21    {
22        case 1:
23            strong();
24            break;
25        case 2:
26            armstrong();
27            break;
28        case 3:
29            harshad();
30            break;
31        case 4:
32            duck();
33            break;
34
35        default:
36            printf("Invalid choice");
37    }
```

```
File Machine View Input Devices Help
1 2 3 4
Open
37 }
38 return 0;
39 }
40
41 /* Checking if strong number or not */
42
43 void strong()
44 {
45     int x, digit, tvar;
46     int sum = 0;
47     int fact, i;
48
49     printf("Enter a positive integer:");
50     scanf("%d", &x);
51     tvar = x;
52
53     while(tvar>0)
54     {
55         digit = tvar%10;
56         fact = 1;
57         for(i = 1; i<= digit; i++)
58         {
59             fact = fact*i;
60         }
61         sum = sum+fact;
62         tvar = tvar/10;
63     }
64     if(sum == x)
65         printf("%d is a Strong number", x);
66     else
67         printf("\n%d is not a strong number", x);
68 }
69
70 /* Checking if armstrong number or not */
71
72 void armstrong()
```

```
70 /* Checking if armstrong number or not */
71
72 void armstrong()
73 {
74     int a, tvar, digit;
75     int sum = 0;
76     printf("Enter a 3 digit postive integer:");
77     scanf("%d", &a);
78
79     tvar = a;
80     while(tvar>0)
81     {
82         digit = tvar%10;
83         sum = sum+(digit*digit*digit);
84         tvar = tvar/10;
85     }
86     if(sum == a)
87     {
88         printf("It is armstrong number:");
89     }
90     else
91     {
92         printf("Its is not armstrong number:");
93     }
94 }
95 /* Checking if Harshad NUmber or not */
96 void harshad()
97 {
98     int b, tvar, digit;
99     int sum = 0;
100     printf("Enter a number:");
101     scanf("%d", &b);
102
103     tvar = b;
104     while(tvar>0)
105     {
106         digit = tvar%10;
```

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

1 2 3 4

Open

103 tvar = b;
104 while(tvar>0)
105 {
106     digit = tvar%10;
107     tvar = tvar/10;
108     sum = sum + digit;
109 }
110 if(b%sum == 0)
111 {
112     printf("%d is harshad", b);
113 }
114 else
115 {
116     printf("%d is not harshad", b);
117 }
118 }
119
120 /* Checking if duck number or not */
121 void duck()
122 {
123     int tvar, n, digit;
124     int check=0;
125     printf("Enter a number:");
126     scanf("%d", &n);
127
128     tvar = n;
129     while(tvar>0)
130     {
131         digit = tvar%10;
132         if(digit == 0)
133         {
134             check = 1;
135         }
136     }
137     tvar = tvar/10;
138 }
139 if(check == 1)
140 {
141     printf("%d is a duck number", n);
142 }
143 else
144 {
145     printf("%d is not a duck number", n);
146 }
147 }
148 }
```

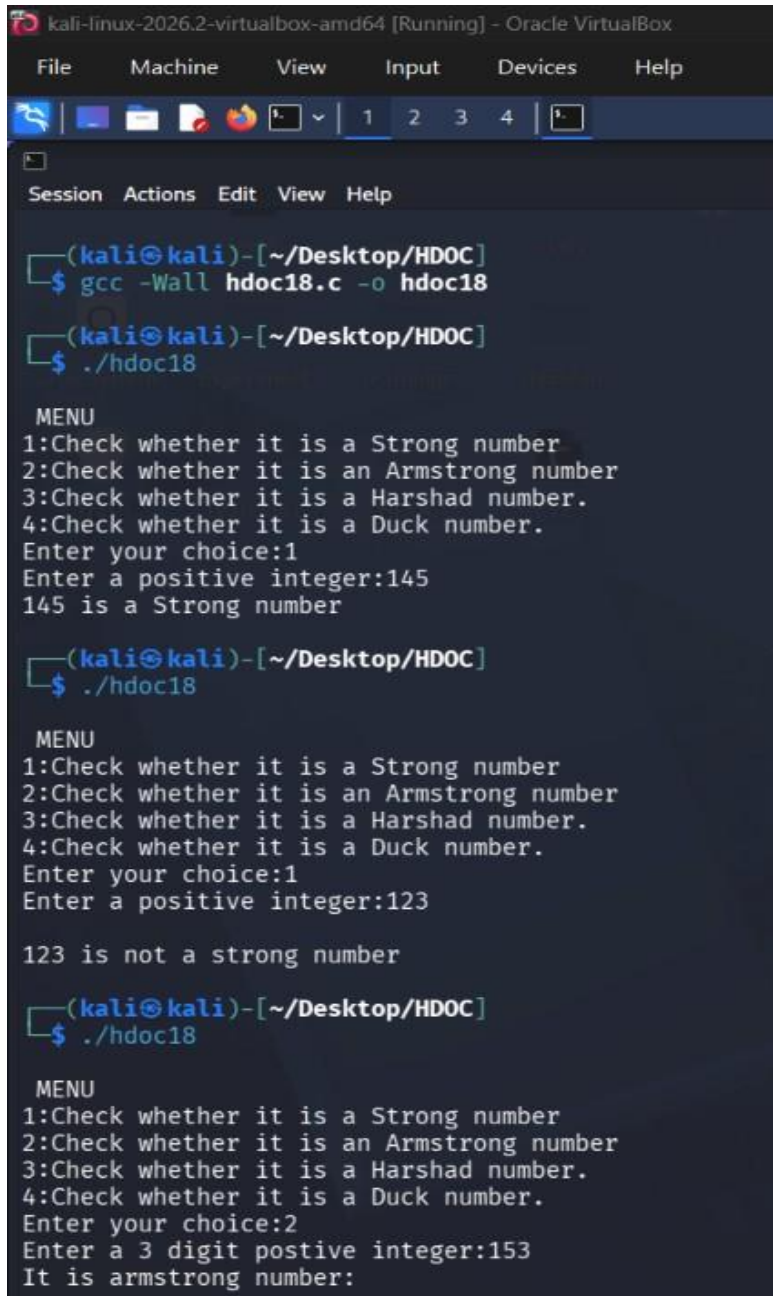

Compile the program using:

```
gcc -Wall hdoc18.c -o hdoc18
```

Execute the code using:

```
./hdoc18
```

Here are some screenshots of the compilation and execution:



```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)-[~/Desktop/HDOC]
$ gcc -Wall hdoc18.c -o hdoc18

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc18

MENU
1:Check whether it is a Strong number
2:Check whether it is an Armstrong number
3:Check whether it is a Harshad number.
4:Check whether it is a Duck number.
Enter your choice:1
Enter a positive integer:145
145 is a Strong number

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc18

MENU
1:Check whether it is a Strong number
2:Check whether it is an Armstrong number
3:Check whether it is a Harshad number.
4:Check whether it is a Duck number.
Enter your choice:1
Enter a positive integer:123
123 is not a strong number

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc18

MENU
1:Check whether it is a Strong number
2:Check whether it is an Armstrong number
3:Check whether it is a Harshad number.
4:Check whether it is a Duck number.
Enter your choice:2
Enter a 3 digit postive integer:153
It is armstrong number:
```

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

It is armstrong number:

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc18

MENU
1:Check whether it is a Strong number
2:Check whether it is an Armstrong number
3:Check whether it is a Harshad number.
4:Check whether it is a Duck number.
Enter your choice:2
Enter a 3 digit positive integer:123
Its is not armstrong number:

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc18

MENU
1:Check whether it is a Strong number
2:Check whether it is an Armstrong number
3:Check whether it is a Harshad number.
4:Check whether it is a Duck number.
Enter your choice:3
Enter a number:18
18 is harshad

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc18

MENU
1:Check whether it is a Strong number
2:Check whether it is an Armstrong number
3:Check whether it is a Harshad number.
4:Check whether it is a Duck number.
Enter your choice:3
Enter a number:19
19 is not harshad

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc18
```

```
(kali㉿kali)-[~/Desktop/HDOC]  
$ ./hdoc18
```

MENU

- 1:Check whether it is a Strong number
- 2:Check whether it is an Armstrong number
- 3:Check whether it is a Harshad number.
- 4:Check whether it is a Duck number.

Enter your choice:5

Invalid choice

```
(kali㉿kali)-[~/Desktop/HDOC]  
$
```

24°C